

Q1. Answer the following:

Q1(a) Define cryptography.

Answer: Cryptography is the technique of protecting information by converting plaintext into ciphertext using an encryption algorithm and a key.

Q1(b) What is the CIA triad?

Answer: The CIA triad consists of **Confidentiality, Integrity, and Availability**, which are the three basic principles of information security.

Q1(c) Name any two classical encryption techniques.

Answer: Two classical encryption techniques are **Caesar Cipher** and **Vigenère Cipher**.

Q1(d) Expand DES and AES.

Answer:

DES: Data Encryption Standard

AES: Advanced Encryption Standard

Q1(e) What is the purpose of the Feistel structure in block ciphers?

Answer: The Feistel structure divides a block into two halves and processes them through multiple rounds to provide **confusion and diffusion** in encryption.

Q2. Describe the CIA Triad with Suitable Real-World Examples.

- The **CIA Triad** is the basic security model used in Cyber Security and Cryptography. It helps protect information from unauthorized access, modification, and loss. The three main principles of the CIA Triad are:
 - **C – Confidentiality**
 - **I – Integrity**
 - **A – Availability**
- These principles ensure that information remains secure and accessible only to authorized users.

1. Confidentiality: Confidentiality means that information should be accessible only to authorized users. It prevents unauthorized people from viewing sensitive data.

Example: Only the account holder can view their online banking details using a username and password.

2. Integrity: Integrity means that the data should not be changed or modified without permission. It ensures that the information remains accurate and trustworthy.

Example: A student's examination marks cannot be changed by unauthorized users in the university database.

3. Availability: Availability means that information and services should be available whenever authorized users need them. The system should work properly without unnecessary downtime.

Example: An ATM service should be available 24×7 so that customers can withdraw money at any time.

Advantages of CIA Triad

- Protects sensitive information from unauthorized access.
- Ensures data remains accurate and trustworthy.
- Keeps systems and services available whenever needed.

Q3. Encrypt the plaintext "NETWORKSECURITY" using the Vigenère Cipher with the keyword "ATTACK". Show the complete encryption process by repeating the key, converting characters to their corresponding numerical values (A = 0, B = 1, ..., Z = 25), performing the encryption using modulo 26, and write the final ciphertext. [5 Marks]

Plaintext: NETWORKSECURITY

Key: ATTACKATTACKATT

Position	Plaintext	P	Key	K	(P+K) mod 26	Cipher
1	N	13	A	0	13	N
2	E	4	T	19	23	X
3	T	19	T	19	$38 \bmod 26 = 12$	M
4	W	22	A	0	22	W
5	O	14	C	2	16	Q
6	R	17	K	10	$27 \bmod 26 = 1$	B
7	K	10	A	0	10	K
8	S	18	T	19	$37 \bmod 26 = 11$	L
9	E	4	T	19	23	X
10	C	2	A	0	2	C
11	U	20	C	2	22	W
12	R	17	K	10	$27 \bmod 26 = 1$	B
13	I	8	A	0	8	I
14	T	19	T	19	$38 \bmod 26 = 12$	M
15	Y	24	T	19	$43 \bmod 26 = 17$	R

Plaintext : NETWORKSECURITY

Key : ATTACKATTACKATT

Ciphertext: NXMWQBKLXCWBIMR

Q4. Encrypt the plaintext "CRYPTOGRAPHY" using the Playfair cipher with key "PPSAVANIUNIVERSITY". Show all steps. [5 Marks]

PLAINTEXT : CRYPTOGRAPHY

KEY : PPSAVANIUNIVERSITY

Unique letters from the key: **P S A V N I U E R T Y**

P S A V N

I U E R T

Y B C D F

G H K L M

O Q W X Z

Plaintext: CR YP TO GR AP HY

Pair 1: C → D , R → E

Pair 2: Y → G , P → I

Pair 3: T → I , O → Z

Pair 4: G → R , R → L

Pair 5: A → V , P → S

Pair 6: H → G , Y → B

Ciphertext: DEGIIZLIVSGB

Q5. Explain the Feistel structure of DES with a neat block diagram. [5 Marks]

- **DES (Data Encryption Standard)** is a **symmetric-key encryption algorithm** used to secure digital data. It uses the **same secret key** for both encryption and decryption. DES encrypts data in **64-bit (8-byte) blocks** using a **56-bit secret key** and performs **16 rounds** of encryption based on the **Feistel Structure**.
- Although DES is now replaced by AES because of its smaller key size, it is still studied to understand the basics of cryptography.

Steps of DES Encryption

- Plaintext (64 bits) is given as input.
- Apply Initial Permutation (IP).
- Split the block into Left half (L_0) and Right half (R_0), each of 32 bits.
- Perform 16 Feistel rounds.
- In each round: - Expand R from 32 bits to 48 bits - XOR with the Round Key - Pass through S-Boxes -
- Apply P-Box permutation - XOR with L
- After the 16th round, swap the two halves.
- Apply Final Permutation (FP) to get the Ciphertext.

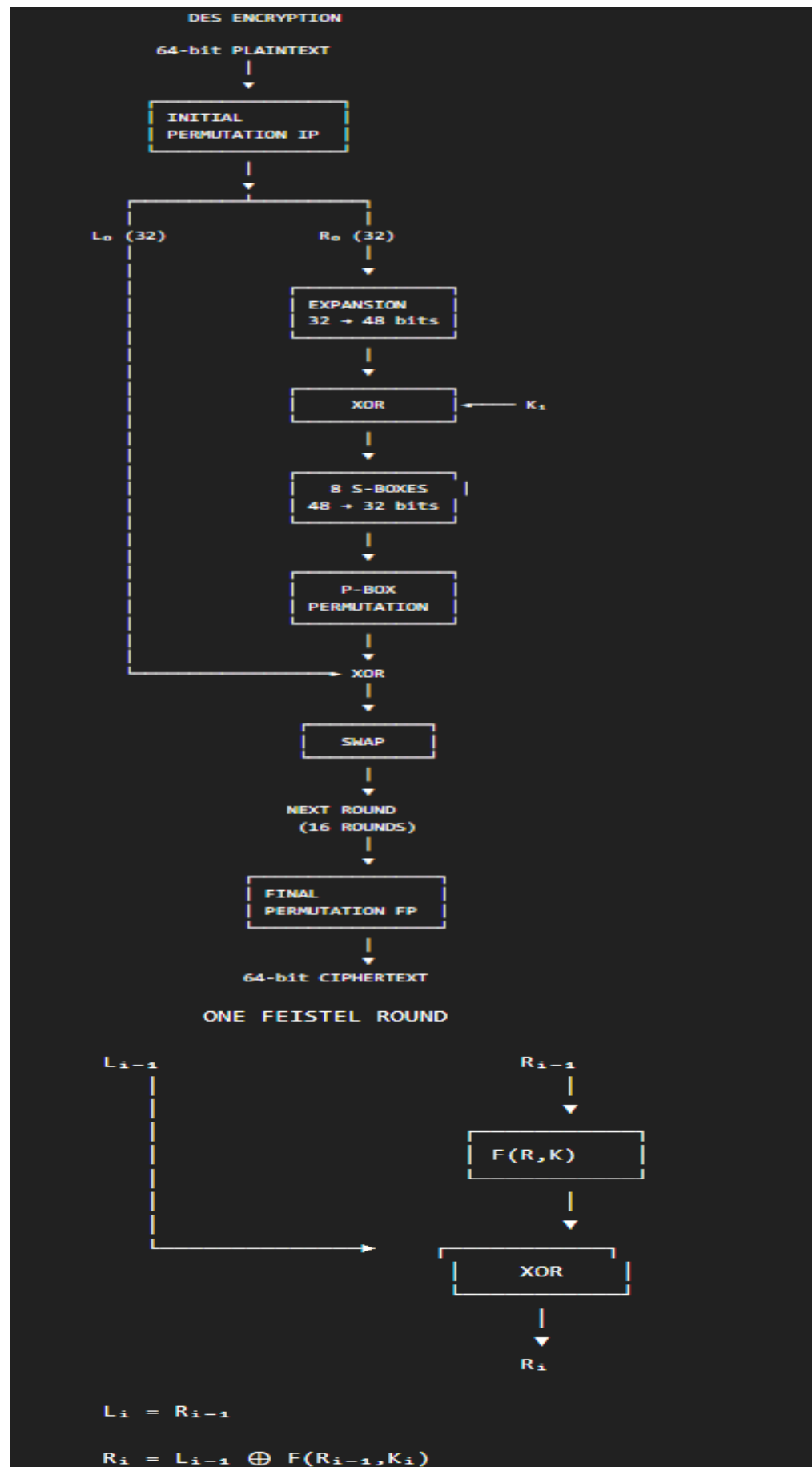
Feistel Structure

In a Feistel structure, the plaintext block is divided into two equal halves: **Left (L)** and **Right (R)**. During each round, one half is processed using a round function and a round key, and then combined with the other half using XOR.

The important equations are:

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$



Q6. Explain Shannon's principles of confusion and diffusion. How are they achieved in modern block ciphers? [5 Marks]

Shannon's Principles of Confusion and Diffusion

In **1949**, **Claude Shannon** introduced two fundamental principles for designing secure cryptographic systems: **Confusion** and **Diffusion**. These principles are used in modern encryption algorithms such as **DES** and **AES**.

1. Confusion

Confusion means making the relationship between the **plaintext**, **ciphertext**, and **encryption key** as complex and unpredictable as possible.

Purpose of Confusion:

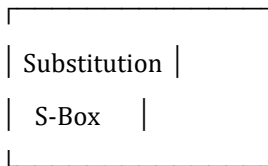
- Hides the relationship between the key and ciphertext.
- Makes it difficult for an attacker to determine the encryption key.
- Makes the effect of the key complex.

How is Confusion achieved?

Confusion is mainly achieved through **substitution operations**, especially **S-Boxes**.

Plaintext

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Ciphertext

2. Diffusion

Diffusion means spreading the influence of each plaintext bit across many ciphertext bits.

Purpose of Diffusion:

- Hides statistical patterns in the plaintext.
- Ensures that changing one plaintext bit affects many ciphertext bits.
- Helps prevent frequency and statistical analysis attacks.

How is Diffusion achieved?

Diffusion is mainly achieved using **Permutation operations**, such as **P-Boxes**, together with multiple encryption rounds.

Plaintext

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Permutation

P-Box

|



Ciphertext

Confusion vs Diffusion

Feature	Confusion	Diffusion
Purpose	Hides relationship between key and ciphertext	Spreads plaintext information
Achieved by	Substitution / S-Boxes	Permutation / P-Boxes
Main focus	Key complexity	Plaintext complexity
Protects against	Key analysis	Statistical and frequency analysis
Effect	Makes key difficult to determine	Makes plaintext patterns disappear